

efficient machine code in the background, while interpreting the JavaScript in the foreground. Once compilation is finished, Chakra switches to the compiled and significantly faster machine code optimized for that PC. Chakra has been systematically tuned to focus on patterns that are found in real world sites.

Although not a goal for the performance work on Chakra, background compilation and many other improvements mean that Internet Explorer 9 scores very well in popular JavaScript benchmarks like WebKit's SunSpider. According to WebKit's SunSpider test, IE9 is the fastest JavaScript browser as of the Internet Explorer 9's release, when the benchmark was last used. However, measuring the performance of a browser by looking at small parts of individual subsystems is inherently flawed. Instead, better benchmarks represent real coding patterns and engage multiple subsystems at once.

New layout system

Internet Explorer 9 has an all new layout system that is optimized to reduce memory use, particularly for HTML5 websites, which will place even more demands on browsers. We find that for large and visually complex websites, the new layout engine reduced memory uses in some cases by as much as 50 percent. In addition, the new layout system forms the foundation for the graphically rich HTML5 applications of tomorrow to ensure that HTML5 capabilities such as SVG are not only fast, but are also able to scale to the needs of developers.

Add-on Performance Advisor

An addition to the feature in Internet Explorer 8 that exposed add-on load time, Add-on Performance Advisor notifies you when add-ons are slowing down your browsing session. You are notified if the total load time of all enabled add-ons takes more than 0.2 seconds, giving you an opportunity to make an informed decision to use the add-ons that you find valuable and disable those that are less useful or too slow.

Hang Recovery

New in Internet Explorer 9, this feature isolates the impact of a hung tab to the individual tab, so that other tabs and the overall browser continue to operate. When a website hangs because of a long running script or other operation, it causes your browser to become nonresponsive. Hang recovery in Internet Explorer 9 means you can continue browsing on